

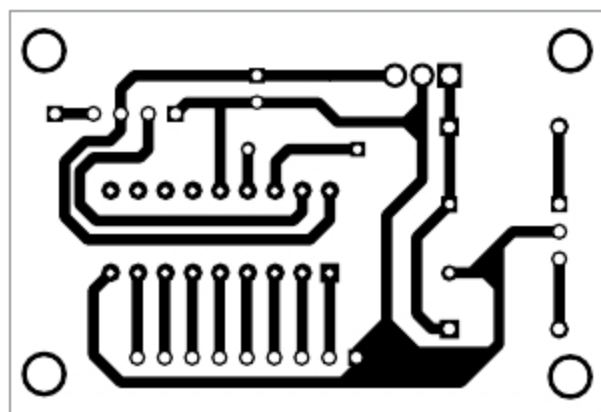


Fig. 6: Actual-size PCB layout of the transmitter unit

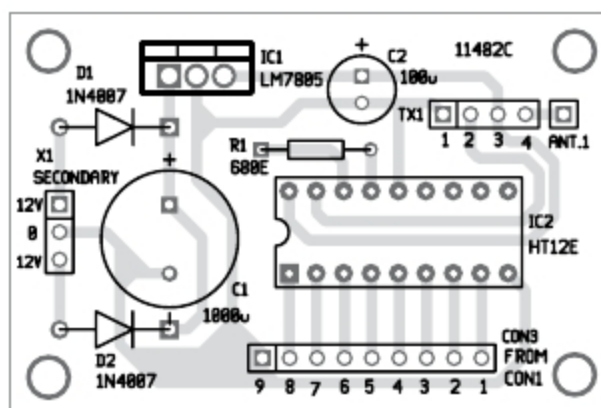


Fig. 7: Components layout for the transmitter unit PCB

Water continues to fill the tank from the level designated by S1. With water level increasing towards the top of the tank, output of the sensors traverses the path, as indicated in Table 1. In doing so, whenever S6 becomes logic low, the corresponding scanned address at the receiver counter 74LS164 output Q6 of IC7 goes low.

Logic state of Q6 of IC7 is inverted by gate N4 of IC4 and its output at pin 11 is fed to input pin 10 of gate N3. The other input gate of N3 is fed with VT (which is at logic high now). Consequently, its output (pin 8) goes high to logic low, resetting the IC9 and, hence, de-energises the relay. The pump stops.

Construction and testing

An actual-size PCB layout of the transmitter unit is shown in Fig. 6 and its components layout in Fig. 7. Assemble the circuit on the PCB and connect 230V AC to the primary of X1.

Similarly, an actual-size PCB layout of the receiver unit is shown in Fig. 8 and its components layout in Fig. 9. Assemble the circuit on the PCB and connect 230V AC to the primary of X2.

Relay RL1 is not connected on the PCB; only relay coil terminals be connected on the PCB. The motor, fuse and AC mains also need to be connected externally.

LED3 through LED10 are used to indicate the water level of the tank.

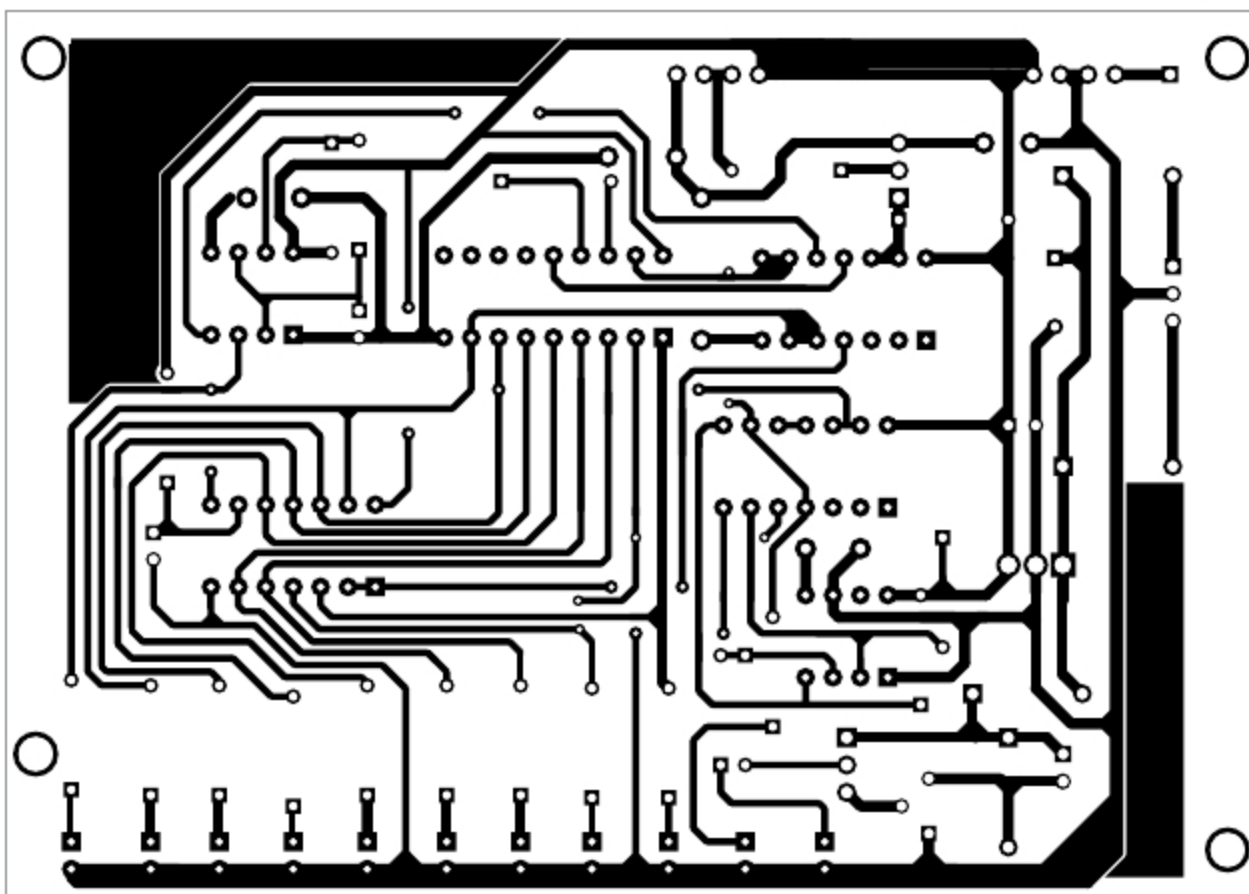


Fig. 8: Actual-size PCB layout of the receiver unit

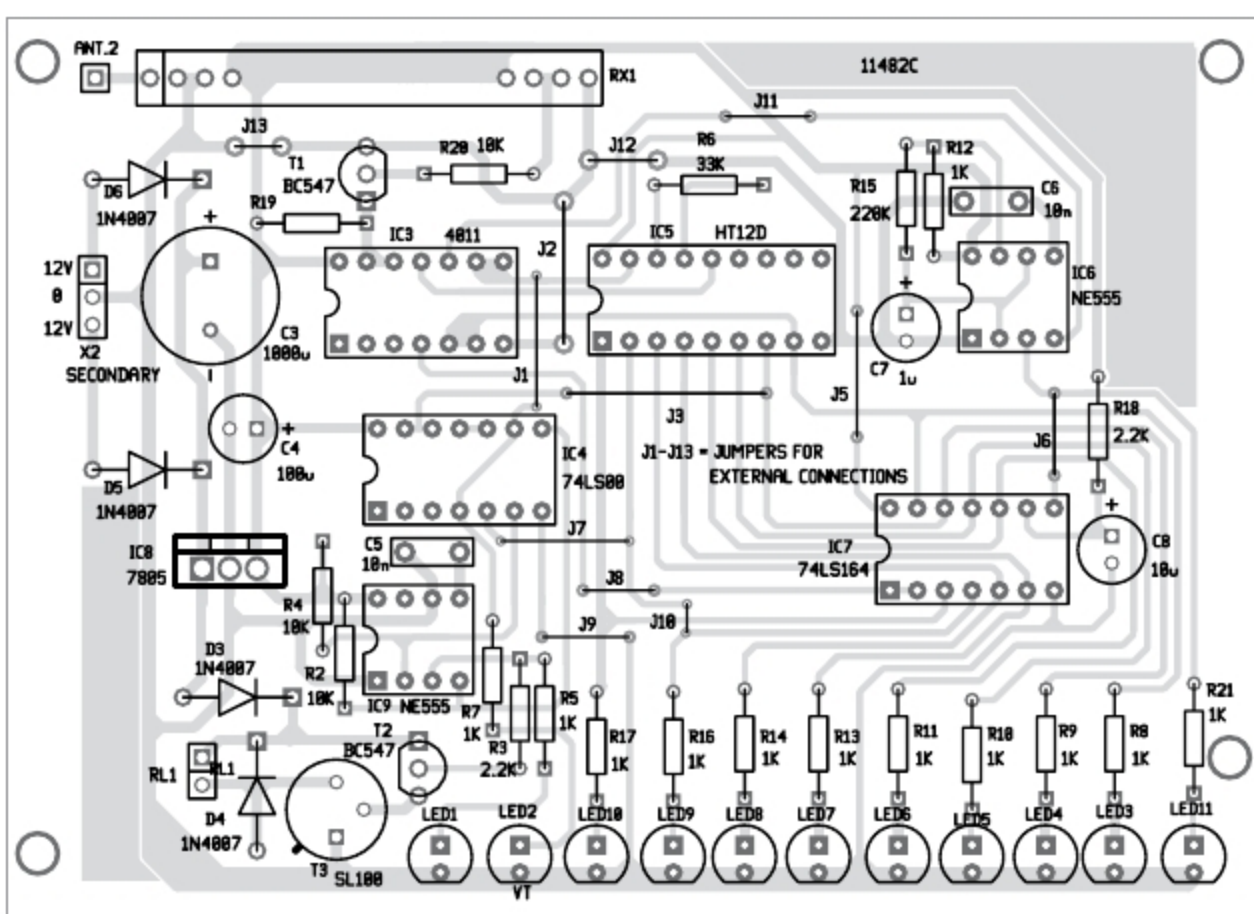


Fig. 9: Components layout for the receiver unit PCB

LED3 is for maximum, while LED10 is for minimum-level indications. Glowing of LEDs means the tank is empty.

Assemble two separate units, one for the transmitter and the other for the receiver. To construct the sensor assembly, metal clamps may be used as shown in Fig. 4. These are placed over an insulated aluminium pipe. PVC pipe sleeve is used for insulation. However, the metal pipe can be replaced with a hard plastic pipe, in

which case no sleeve is required.

Insulated wires are used to connect the sensors to encoder address lines (A0 to A7) of HT12E. To reduce the length of these wires, the transmitter unit may be mounted in a water-tight plastic box over the tank. **EFY**



Arup Kumar Sen is ex-technical officer at SAIF Bose Institute, Kolkata